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Highlights

- Sole-author paper in *Finance Research Letters* (Scopus D1)
- Oral presentations at VPH 2026 and CMFF 2025
- 1st Place, EELISA international competition
- Two 1st places at BME research conferences

Skills

ML & Data
PyTorch, graph neural networks, image segmentation, Bayesian methods, SQL

UQ & Simulation
Polynomial chaos expansion, Gaussian processes, Dakota, VTK, PyVista

Programming
Python, MATLAB, C, C++, R, TypeScript, Mathematica, Solidity

Engineering
AutoCAD, Inventor, LabVIEW, MPLAB

Tools
Git, Linux, Slurm (HPC), Docker, Jupyter, LaTeX

Languages

Hungarian: native
English: C1 (IELTS 8.0)
German: B2
French: B1

Interests

Ultra-endurance cycling
Drumming
PC hardware & overclocking
DIY: built a garden house

Damján MIKLÓS

Mechatronics Engineering BSc student

Machine learning and uncertainty quantification for cardiovascular simulations

Education

BSc in Mechatronics Engineering Sep 2023 – Jan 2027 (expected)
Budapest University of Technology and Economics (BME)
GPA: **4.71/5** · Ranked **#1 of 199** in Mechatronics by ECTS-adjusted GPA
246 ECTS (210 required) · Advanced Mathematics Track

Guest Student, ELTE Faculty of Science (Fourier Analysis university lectures during high school) 2023

High School Diploma, ELTE Trefort Ágoston High School 2017–2023

Research

Uncertainty Quantification for Aneurysm Hemodynamics Nov 2024 – present
Research Assistant, BME Department of Hydrodynamic Systems

- Developed an uncertainty quantification framework for aneurysm blood-flow simulations with polynomial chaos expansion (PCE) and Gaussian-process surrogate models (Dakota, Python, MATLAB)
- Derived a new semi-analytical method that constructs orthogonal PCE polynomials for truncated, correlated normal input distributions (unpublished)
- Built an efficient virtual patient sampling algorithm in MATLAB

- **Oral presentations** at VPH 2026 (Virtual Physiological Human), Milan, and CMFF 2025, Budapest · part of GEMINI
- **1st Place**, EELISA Scientific Competition 2026, Madrid · Biomedical section
- **1st Place**, GHK Special Award, BME Scientific Students' Conference (TDK) 2025 · Fluid Dynamics section

Generative Graph Neural Network for Carotid Artery Anatomies 2025 – present
BSc thesis, BME Department of Hydrodynamic Systems

- Developed a conditional generative graph neural network that creates anatomically accurate internal carotid artery (ICA) meshes with aneurysms from a single condition vector
- Built the cleaning and data processing pipeline for high-resolution ICA meshes in Python (VTK, PyVista)
- Trained on Komondor, Hungary's national supercomputer, under a successfully awarded computing grant to my research.
- Developing a neural network for segmenting calcified ICAs

- **xxx Place**, BME TDK 2026 · Neural Networks section

Irrationality in Skewness Preferences 2023 – 2026
Independent research, BME Department of Finance

- Built a quasi-Monte Carlo simulation of hypothetical skewed return distributions, deriving a model-independent upper bound on the skewness premium from tail-risk measures
- Showed that the premia reported in six empirical studies exceed this bound at realistic horizons, pointing to behavioural drivers; code released open source

- **Published:** Miklós, D. (2026). *Finance Research Letters*, 109, 110514. doi:10.1016/j.frl.2026.110514 – sole author, Scopus D1 journal
- **1st Place**, Morgan Stanley Award, BME TDK 2024 · Finance section
- **3rd Place**, National Scientific Students' Conference (OTDK) 2025

Other Experience & Projects

Online Math & Physics Tutor Superprof 2023 – present
2000+ hours of university and high school-level tutoring in English and Hungarian; among the top-rated tutors nationwide

DynamicSpeed for YouTube 2026
Browser extension for Chrome and Firefox
Matches playback speed to a target words-per-minute estimated from captions